TECHNICAL ROADMAP & ARCHITECTURE DOCUMENT

Sovereign Conclave v0.2.0

From Claude Code skill to a live, cryptographically verifiable multi-agent deliberation protocol. Content-addressed transcripts, ML-DSA signed verdicts, a structured appeals route, and a token-economy architecture targeting 70–90% cost reduction.

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00 Executive Summary

Sovereign Conclave v0.1 is a working proof of concept: a 21-seat council of historically inspired personas, organized into four tiers (Generals, Sovereigns, Justices, Inner Sanctum), deliberating hard architectural and institutional decisions inside Claude Code, with the Marshall verifier seat grounding claims against a frozen Evidence Ledger. It proves the deliberation model. It does not yet prove the *protocol*. Three gaps separate v0.1 from something that can run live, in public, on questions that matter:

- **No verifiability.** Transcripts are plain markdown. Nothing prevents post-hoc editing of a verdict, a round, or the Evidence Ledger itself. For the eSiasa civic-governance use case — public, auditable deliberation over policy questions — this is disqualifying.
- **No recourse.** A verdict is final by construction. There is no dissent registration, no minority report, no structured trigger for re-deliberation when new evidence lands or procedure was violated.
- **Naive token economics.** Every persona sees the full transcript every round. Cost grows quadratically with rounds and linearly with seats. A full 21-seat, 4-round deliberation on a frontier model costs an order of magnitude more than it needs to.

v0.2.0 closes all three. The verifiability layer content-addresses every artifact (CIDv1 over canonical CBOR), Merkle-chains rounds into a DAG, and signs verdicts and per-round attestations with ML-DSA-65 via your own `pqcrypto` package — making the Conclave one of the first deliberation systems with post-quantum-signed outputs, and dogfooding your FIPS 204 implementation in the process. The appeals route adopts a Kleros-inspired escalation ladder (re-deliberation → adversarial appellate panel → human arbiter) with dissents and appeals referencing the content-hashed record of the original proceeding. The token architecture combines prompt caching, the Batch API's 50% discount, model cascades, structured distillation, and sparse topologies to cut projected per-deliberation cost by 70–90% against the v0.1 baseline, with worked math in Section 04.

The delivery vehicle stays dual-track: the Claude Code skill remains the reference implementation and fastest iteration surface, while a Serverpod service (`conclave_server`) becomes the live protocol host — streaming deliberation progress over WebSockets, persisting the DAG in Postgres, holding live session state in Redis, and exposing the API surface eSiasa will consume.

> **THE ONE-SENTENCE PITCH**
>
> v0.2.0 turns "an AI council wrote this verdict" into "an AI council wrote this verdict, here is the tamper-evident, PQ-signed, replayable proof of exactly how, and here is how you appeal it."

01 Current-State Analysis: v0.1 Codebase

1.1 What exists

The repository is a Claude Code skill of roughly twelve files: a top-level `SKILL.md` orchestrator, persona definitions for the 21 seats across the four tiers, the Marshall verifier specification, the Evidence Ledger convention (evidence is frozen before round one; personas may cite only ledger entries), round orchestration instructions (opening statements → challenge rounds → synthesis → verdict), and output templates for the final verdict document. Deliberation state lives entirely in the conversation context and the markdown files the skill emits.

1.2 What works well

- **The persona tier structure is genuinely good design.** Separating Generals (execution pragmatists), Sovereigns (strategic horizon), Justices (constraint and principle enforcement), and the Inner Sanctum (synthesis) creates productive friction instead of the mode-collapse that plagues naive multi-agent debate, where all agents converge to the same voice within two rounds.
- **The Marshall + frozen Evidence Ledger is the single most important architectural decision in v0.1.** Grounding claims against a frozen corpus is exactly what the LLM-as-judge literature says you need to combat sycophantic drift and fabricated citation. It also happens to be the natural anchor point for content hashing: the ledger is already conceptually immutable, so making it cryptographically immutable is a small step.
- **Skill packaging keeps iteration cost near zero.** Persona edits are markdown edits. No deploy cycle.

1.3 Structural limitations to fix in v0.2.0

Limitation	Consequence	v0.2.0 answer
Plain-markdown transcripts, mutable after the fact	Zero tamper evidence; unusable for public/civic deliberation	Content addressing + Merkle round DAG + ML-DSA signatures (§02)
Verdict finality with no recourse	Errors and procedural violations are unrecoverable	Dissent registration + appeals ladder (§03)
Full transcript replayed to every persona, every round	O(seats × rounds²) token growth; ~$15–25 per full deliberation at frontier pricing	Caching, cascades, distillation, sparse topology (§04)
State lives only in conversation context	No resume, no replay, no concurrent deliberations, context-window ceiling caps deliberation depth	Persisted round DAG in Postgres; context reconstructed from distilled summaries (§05)
No telemetry	Cost and quality are unmeasurable; no basis for optimizing	Per-round token/cost/latency telemetry (§06)
Persona definitions are unversioned prose	Cannot attribute a verdict to a specific persona version; silent drift	Signed, content-addressed persona manifests (§02.5, §06)

HONEST CAVEAT

The single-context design also has a subtle quality problem: all 21 personas are role-played by one model instance in one context, so cross-persona information leakage is total. A persona "knows" what another persona is about to argue because the same forward pass produced both. v0.2.0's per-persona API calls (required for caching and cascades anyway) also fix this epistemic hygiene issue — genuine information asymmetry between seats becomes possible for the first time.

02 Cryptographic Verifiability Layer

This is the flagship of v0.2.0 and the layer that plays directly to your stack. Design goal: every artifact a deliberation produces — the question, the evidence ledger, every persona turn, every round, every vote, the verdict, every dissent, every appeal — is content-addressed, hash-linked into a DAG, and attested with ML-DSA-65 signatures produced by `pqcrypto`. Verification requires no trust in the Conclave operator: an independent verifier with the artifact set and public keys can recompute every hash and check every signature offline.

2.1 Canonical serialization first, or nothing else works

Hashes are only meaningful over canonical bytes. Two serializations of the same logical object must produce identical digests, or the whole scheme collapses into "hash of whatever the serializer happened to emit that day." Two viable standards:

- **RFC 8785 JCS (JSON Canonicalization Scheme):** deterministic member ordering, ECMAScript number serialization, UTF-8. Pro: human-readable artifacts, trivial to audit by eye. Con: JSON number edge cases (i64 precision beyond 2^{53}) and string escaping subtleties are exactly the kind of thing that silently breaks cross-implementation hashing.
- **Deterministic CBOR (RFC 8949 §4.2 core deterministic encoding, the profile CTAP2 uses):** shortest-form integers, definite lengths, bytewise-sorted map keys. Pro: compact, no floating-point ambiguity if you ban floats (do this — the schema needs no floats; represent costs as integer micro-units), and it is the same canonical form the IPLD ecosystem (dag-cbor) already standardizes.

> **DECISION**
>
> Use **dag-cbor** (deterministic CBOR per the IPLD spec) as the canonical form, with CIDv1 (multihash sha2-256, dag-cbor codec) as the artifact identifier. This buys IPFS-compatibility for free — any Conclave artifact can later be pinned to IPFS and its CID will match without re-encoding — and sidesteps every JSON canonicalization footgun. Emit a JCS-JSON *view* alongside for human audit, clearly labeled non-normative.

2.2 The artifact model and round DAG

Every object is an immutable node identified by its CID. Links are CIDs, so the structure is a Merkle DAG by construction — tampering with any node changes its CID and breaks every link pointing at it, transitively invalidating the verdict's root hash.

```
DeliberationManifest          // root object, the "case number"
 ├─ question: CID → Question
 ├─ evidenceLedger: CID → EvidenceLedger
 │    └─ entries: [CID → EvidenceEntry]     // each entry hashes source bytes
 ├─ panel: CID → PanelManifest
 │    └─ seats: [CID → PersonaManifest@version]
 ├─ protocolParams: CID → ProtocolParams    // rounds, quorum, vote weights
 └─ rounds: [CID → Round]                   // append-only chain

Round {
  index: uint,
  prev: CID → Round | DeliberationManifest, // hash chain
  turns: [CID → Turn],
  distilledSummary: CID → Summary,          // the ONLY carried-forward context (§04)
  attestation: MLDSASignature               // orchestrator signs the round CID
}

Turn {
  persona: CID → PersonaManifest,
  inputContextHash: CID,                    // exactly what the model saw
  modelId: string, samplingParams: {...}, seed: uint?,
  output: CID → Content,
  citations: [CID → EvidenceEntry]          // claims must link ledger CIDs
}

Verdict {
  round: CID → Round,
  decision: Content, votes: [Vote],
  dissents: [CID → Dissent],                // §03
  signature: MLDSASignature                 // Conclave identity key signs verdict CID
}
```

Key properties: **(a)** the Evidence Ledger's freeze becomes cryptographic — its CID is fixed in the manifest before round one, so late evidence injection is detectable by anyone; **(b)** `Turn.citations` are CID links, so "cite only the ledger" is now machine-checkable, and the Marshall's grounding job becomes verifying link validity plus semantic faithfulness rather than policing free-text citations; **(c)** `inputContextHash` pins exactly what each persona saw, which is what makes the replay and appeals machinery honest.

2.3 Signatures: ML-DSA-65 via pqcrypto

Three key roles, all ML-DSA-65 (FIPS 204 security category 3 — the right default; ML-DSA-44 saves bytes but category 2 is a needless haircut for long-lived civic records, and ML-DSA-87's size cost buys nothing here):

Key	Signs	Lifecycle
Conclave identity key	Verdicts, appeal outcomes, transparency-log checkpoints	Long-lived, HSM/KMS-held in production, published fingerprint
Session orchestrator key	Per-round attestations, session manifest	Ephemeral per deliberation; cert chain: identity key signs (session pubkey + manifest CID)
Persona registry key	PersonaManifest versions (§06)	Long-lived; enables third-party persona authorship later

Signature payloads are always CIDs (32-byte digests wrapped in multihash framing), never raw documents — standard hash-then-sign, and ML-DSA-65 signatures at ~3,309 bytes stay constant

regardless of artifact size. This is also a live production workout for `pqcrypto`'s ML-DSA path: real key lifecycle, real cert-chain semantics, real verification tooling — considerably better dogfood than KAT vectors.

> **EDGE CASE: RETRIED ROUNDS**
>
> LLM calls fail. If round *n* is retried, the failed attempt must not simply vanish — that is silent history editing. Model retries explicitly: a Round may carry `supersedes: CID → Round` pointing at an aborted attempt, and aborted rounds remain in the DAG marked `status: aborted` with the failure reason. The chain stays append-only; nothing is ever deleted. The same mechanism handles a persona turn that violates protocol (e.g., cites outside the ledger) and is struck: the struck turn stays, marked, with the Marshall's objection linked.

2.4 Transparency log

Hash chains prove internal consistency but not *existence at a point in time*, and a malicious operator could maintain forked histories showing different verdicts to different audiences (a split-view attack). The fix is the Certificate Transparency / Sigstore Rekor pattern: an append-only Merkle tree (RFC 6962 structure) over verdict CIDs, with signed tree heads published periodically. Inclusion proofs let anyone verify a verdict is in the log; consistency proofs let monitors verify the log never rewrote history. v0.2.0 scope: run your own single-operator log inside `conclave_server` (a Postgres table plus RFC 6962 Merkle tree math is a weekend of Dart, and you already have the hashing primitives). v0.3+ options: cross-publish tree heads to an independent witness — even a signed tweet-sized checkpoint on a public timeline breaks split-view — or anchor into Rekor itself. Do not touch blockchains for this; a transparency log gives you the auditability without the circus.

2.5 Reproducibility: what to claim and what not to claim

Be precise here, because overclaiming reproducibility is how verifiability systems lose credibility. LLM inference is not bit-reproducible across time: providers update models, batch non-determinism exists even at temperature 0, and seeds are not honored across all APIs. So the honest claim hierarchy is:

1. **Tier 1 — Record integrity (always):** the transcript you are reading is exactly what was produced, unmodified. This is what the DAG + signatures give you, and it is the claim that matters for audit.
2. **Tier 2 — Input reproducibility (always):** `inputContextHash` + `modelId` + `samplingParams` + `seed?` pin exactly what was asked of which model, how. Anyone can re-run the same input and compare outputs.
3. **Tier 3 — Output reproducibility (never promise):** re-running may produce different text. Do not claim otherwise anywhere in the docs. The appeals route (§03), not re-execution, is the remedy for "the model said something different this time."

03 The Appeals Route

An appeals mechanism is what turns a verdict generator into a deliberation *protocol*. Design sources: appellate court procedure (grounds-based review, not free re-litigation), Kleros's escalating-jury model (each appeal round convenes a larger/different panel at higher stakes), and the AI-safety-via-debate literature (adversarial review as an error-correction mechanism). The v0.1 verdict-is-final model inverts into: *every verdict is final unless a valid appeal trigger fires within the finality window.*

3.1 Dissent registration (the foundation)

Appeals need standing, and standing needs dissent. In the verdict round, every voting persona either joins the majority or files a `Dissent` artifact: a structured minority report with its own CID, linked from the Verdict, citing the specific ledger evidence and prior turns (by CID) the majority allegedly misweighed. Dissents are first-class signed artifacts — they appear in the published verdict document, permanently. This alone raises deliberation quality: personas prompted with "your dissent will be published and hash-linked forever" produce measurably sharper minority reasoning than personas whose disagreement evaporates into a vote tally.

3.2 Appeal triggers (grounds-based, not vibes-based)

Trigger	Standing	Window	Gate
T1: New evidence — material evidence not in the frozen ledger	Any party (human or persona dissent)	Configurable; default 30 days for civic use	Marshall screens: is it genuinely new, genuinely material? Screening verdict is itself a signed artifact.
T2: Procedural violation — protocol breach (uncited claim survived, quorum miscount, ledger tampering)	Anyone — this is the trigger the crypto layer makes *publicly checkable*	No limit for integrity violations	Mechanical verification against the DAG; violations proven by hash/signature check auto-grant the appeal.
T3: Dissent threshold — ≥30% of voting weight dissented	Automatic	Immediate	None; auto-schedules one appellate review.
T4: Stakeholder petition (civic mode) — affected parties petition	Defined stakeholder class	Configurable	Threshold count + human clerk screen.

3.3 The escalation ladder

1. **Tier A — Re-deliberation with expanded context.** Same panel composition (fresh instances), original distilled record + the appeal grounds + admitted new evidence (as a signed *ledger supplement*, preserving the original ledger's CID). Cheapest tier; handles T1 well.
2. **Tier B — Adversarial appellate panel.** Different personas from the original panel (disjoint seat draw — Kleros-style fresh jury), plus a mandated *advocatus diaboli* whose brief is to steelman the dissent. The appellate panel reviews the hashed record; it does not re-litigate from scratch. Its scope is the appeal grounds only, mirroring appellate-court review standards. Output: affirm, reverse, or remand (order a new Tier-A deliberation with corrected procedure).

3. **Tier C — Human arbiter.** Final and mandatory as the terminus. A named human (or human panel, for eSiasa: a designated review board) receives the full verifiable record — original deliberation, dissents, appellate reasoning, all CIDs checkable — and rules. The ruling is signed and closes the case. *No fully-automated finality for civic decisions, ever.* This is not a safety fig leaf; it is what makes the system legible to institutions that will never accept "the AI court's decision is final."

Every appeal artifact (petition, screening decision, appellate verdict, arbiter ruling) links the original `DeliberationManifest` CID and enters the same transparency log, so a case's full procedural history is one connected, verifiable graph. Anti-abuse: appeals per case are capped (default: one per trigger class), Tier B convening requires either a granted T1/T2 or an automatic T3, and repeat frivolous T4 petitions burn the petitioner class's threshold credit — a lightweight cost analog to Kleros's stake-to-appeal without introducing tokens.

04 Token-Efficient Deliberation Architecture

The v0.1 pattern — one giant context, every persona sees everything, every round — is the most expensive possible topology. The v0.2.0 architecture attacks cost on five independent axes that compound. Pricing baseline (verify against current rate cards at implementation time; these move): Opus-class ~$15/$75 per MTok in/out, Sonnet-class ~$3/$15, Haiku-class ~$1/$5. Prompt caching: cache writes ~1.25× base input, cache hits ~0.1× base input. Batch API: 50% off everything, at the cost of async (minutes-scale) turnaround.

4.1 Axis 1 — Prompt caching (the free 60–80% on input)

Restructure every persona call as: `[shared static prefix: protocol rules + evidence ledger + panel roster] + [shared dynamic middle: distilled round summaries] + [persona-specific suffix: persona manifest + turn instruction]`. The static prefix is written to cache once per deliberation and hit by every subsequent call at 0.1× input price. With a 30k-token ledger + protocol prefix and 21 seats × 4 rounds = 84 calls, naive input cost pays for that prefix 84 times; cached, you pay 1.25× once and 0.1× 83 times — a ~92% reduction on the shared-prefix component alone. Cache TTL (5 min default, 1 h extended) comfortably covers intra-deliberation cadence; batch-scheduled rounds should be grouped to stay inside TTL windows or accept re-warming cost, which the math still favors.

4.2 Axis 2 — Structured distillation (kill the quadratic)

After each round, a cheap-model pass emits a `Summary` artifact: positions held, claims made (with citation CIDs), open disagreements, vote-relevant deltas — hard-capped (e.g., 1,200 tokens/round). Subsequent rounds carry *only* the distilled summaries, never raw transcripts. Token growth per round becomes O(1) instead of O(rounds). Full transcripts remain in the DAG for audit and appeals; they just stop riding along in inference context. The summary is content-addressed and its `inputContextHash` proves which raw turns it distilled — so an appeal alleging "the distillation dropped my argument" (a real failure mode: distillation loss compounds across rounds and can systematically erase minority positions) is checkable against the record. Mitigation: the distiller prompt mandates preserving *every* seat's position including minority-of-one, and dissenting personas may flag summary-fidelity objections that force a one-shot re-distill with their turn quoted verbatim.

4.3 Axis 3 — Model cascade

Work	Model tier	Rationale
Vote tallying, formatting, schema validation, summary distillation, consensus detection	Haiku-class (or local — see 4.6)	Mechanical; correctness is checkable
Standard persona turns, opening statements	Sonnet-class	The workhorse; persona fidelity at this tier is empirically sufficient
Inner Sanctum synthesis, verdict drafting, tie-breaks, Tier-B appellate review	Opus-class	Reserve frontier spend for the turns where reasoning depth changes outcomes

Rough split in a 4-round deliberation: ~70% of turns are Sonnet-tier, ~25% mechanical (Haiku/local), ~5% Opus-tier. Cascade escalation is also dynamic: if consensus detection (4.5) flags a round as contentious, its synthesis turn escalates one tier.

4.4 Axis 4 — Sparse topology

The multi-agent debate literature (sparse-communication debate results, DyLAN-style dynamic agent selection, exchange-of-thought topologies) consistently finds that full all-to-all communication is not just expensive but often *worse* — it accelerates convergence to early majority positions. v0.2.0 topology: rounds 1–2 run in **committees** (tier-aligned subsets of 5–7 seats deliberating in parallel), round 3 is a **cross-committee challenge** where each committee's distilled position is attacked by one delegate from each other committee, round 4 convenes the full council on the distilled committee outputs only. Turn count drops from 84 to ~45–55 while *increasing* position diversity in early rounds. Not every persona speaks every round; silent seats still vote, reading the distilled record (a cheap Sonnet read, or free if their committee context is cached).

4.5 Axis 5 — Early exit, memoization, batching

- **Consensus detection:** after each round, a Haiku pass (or embedding cosine over position statements) scores agreement; above threshold with no active dissent flags, skip remaining challenge rounds and go to verdict. Many questions genuinely resolve in two rounds.
- **Deliberation memoization:** content-hash the (question + ledger + protocol params) triple. On collision with a prior case, offer replay-with-adaptation: a single Opus pass reviews the prior verdict against any context delta instead of re-convening. CIDs make the cache key exact and the provenance honest — the new verdict links the precedent's CID, which incidentally bootstraps a citable case-law graph.
- **Batch API:** deliberations are not chat; a round's turns are embarrassingly parallel and latency-tolerant. Run all non-interactive rounds through Batches for a flat 50% discount, stacking with caching. Interactive/live-streamed civic sessions forgo this; scheduled policy deliberations (the eSiasa norm) take it by default.
- **Multi-persona packing (use sparingly):** packing several lightweight persona turns into one structured-output call saves overhead but reintroduces the v0.1 leakage problem — acceptable only for same-committee turns where shared visibility is already the design.

4.6 Local models for the mechanical tier

Tallying, schema validation, canonicalization checks, and first-pass distillation can run on a local Ollama/llama.cpp deployment (e.g., an 8–14B instruct model) at zero marginal token cost. Keep a cloud-Haiku fallback behind the same interface; promote local-model outputs only where a deterministic validator can check them (vote math, schema conformance) or a cheap cloud pass can spot-audit distillations. This tier is also your sovereignty story: the mechanical layer of a civic deliberation can run entirely on Kenyan infrastructure.

4.7 Worked cost model

Configuration	Assumptions	Est. cost/ deliberation
v0.1 naive, frontier model	84 turns, full-transcript context growing to ~60k tokens, Opus-class throughout	~$18–27
+ cascade (Sonnet workhorse)	Same topology, tiered models	~$5–8
+ caching + distillation	Shared prefix cached; O(1) round context	~$2.20–3.50
+ sparse topology + early exit	~50 turns median, 2.8 rounds median	~$1.40–2.20
+ Batch API (scheduled mode)	50% off all of the above	**~$0.70–1.10**

Net: **~92–96% below naive**, and ~70–90% even against a already-sensible cascaded baseline. At ~$1/deliberation, running every contentious county-budget line item through a full verified Conclave becomes an operational rounding error, which is the point: cost architecture is what makes the civic use case real rather than demo-ware. All figures are planning estimates — re-derive against live rate cards before publishing numbers externally.

05 Productionization on Serverpod

5.1 Shape of the system

Dual-track: the Claude Code skill remains the reference protocol implementation and persona-authoring surface; `conclave_server` (Serverpod 3.x, Dart-only monorepo, GCE + Terraform per your existing eSiasa infra decisions) is the live host. Packages:

```
conclave/
 ├─ conclave_core/        // pure Dart: canonical CBOR, CID, DAG, protocol state machine
 │                        // (no Serverpod deps → reusable from the CC skill via CLI)
 ├─ conclave_crypto/      // thin façade over pqcrypto: key mgmt, ML-DSA sign/verify,
 │                        // cert-chain, transparency log Merkle math
 ├─ conclave_server/      // Serverpod app: endpoints, orchestrator, persistence, Redis
 ├─ conclave_client/      // generated client
 └─ conclave_flutter/     // observer/verifier UI (v0.3, stub in v0.2)
```

5.2 Persistence and state

Postgres holds the append-only artifact store: one `artifacts` table (cid PK, canonical bytes, type tag, created_at — never updated, never deleted) plus relational projections (`deliberations`, `rounds`, `turns`, `verdicts`, `appeals`) that are pure indexes over artifact CIDs for query ergonomics; the artifacts are the truth, projections are rebuildable. The transparency log is a `log_entries` table with tree-size-indexed Merkle nodes and a `signed_tree_heads` table. Redis (Memorystore) carries live session state: active deliberation FSM position, cache-warmth timers, per-session token budget counters, and pub/sub fan-out for progress events. Serverpod *future calls* schedule batch-round polling and appeal-window expiry (finality timers) — exactly the feature set that drove your GCE-over-Cloud-Run call, now earning its keep.

5.3 API surface (v0.2.0)

```
// Endpoints (session-authenticated; civic-public reads are unauthenticated)
DeliberationEndpoint:
  Future<CaseRef> convene(Question q, EvidenceLedgerDraft e, ProtocolParams p);
  Stream<DeliberationEvent> watch(CaseRef ref);       // Serverpod stream: round
                                                       // started/turn committed/summary/
                                                       // consensus score/verdict
  Future<VerdictBundle> verdict(CaseRef ref);          // verdict + dissents + proofs
AppealEndpoint:
  Future<AppealRef> petition(CaseRef ref, AppealGrounds g);
  Future<ScreeningResult> screening(AppealRef ref);
VerifyEndpoint (public):
  Future<ArtifactBytes> artifact(Cid cid);
  Future<InclusionProof> prove(Cid verdictCid);
  Future<SignedTreeHead> latestSTH();
```

The `watch` stream is what makes live public deliberation a product: an eSiasa client renders rounds as they commit, each event carrying the CID of the just-committed artifact so the client can verify as it renders. Budget enforcement lives in the orchestrator: every deliberation is convened with a hard token budget; graceful degradation order is (drop to Batch) → (tighten sparse topology) → (early-verdict with reduced rounds, flagged in the manifest) — never silent quality reduction.

5.4 v0.2.0 scope honesty

In scope: the artifact store, orchestrator with all §04 optimizations, signing/verification, transparency log, appeals T1–T3 (T4 stakeholder petitions need eSiasa-side identity work — v0.3), and the verify endpoints. Out of scope for v0.2.0: the Flutter observer UI (stub), multi-tenant persona marketplaces, external witness co-signing, and eSiasa production integration beyond a working API demo against one real policy question.

06 Additional Features

- **Signed persona registry.** PersonaManifests (prompt, tier, vote weight, model-tier floor, version, changelog) are content-addressed and registry-key signed. Verdicts pin exact persona versions; silent persona drift becomes impossible. This is the substrate for third-party persona authorship later — a signed-manifest marketplace is v0.4 territory, but the format ships now.
- **Calibration scoring.** Where deliberations make falsifiable predictions ("this migration will take <3 weeks"), log them with resolution dates; score personas with Brier scores over time. Calibration history can eventually modulate vote weights — ship the telemetry in v0.2.0, defer weight coupling until there is enough data to not be noise-steering.
- **Deliberation replay / time-travel.** Because every turn pins its `inputContextHash`, a replay tool can step through the deliberation exactly as it unfolded, or fork it: "replay from round 2 with this evidence added" convenes a counterfactual deliberation whose manifest links the original — devastatingly useful for appeals prep and for post-mortems on bad verdicts.
- **Deliberation templates.** Pre-baked ProtocolParams + panel compositions for ADR review, RFC review, incident postmortem, procurement evaluation, and (for eSiasa) policy-impact review. Templates are themselves signed artifacts.
- **Evidence provenance.** EvidenceEntry hashes source bytes and records fetch metadata (URL, retrieval time, retriever identity). For civic use, ledger assembly itself becomes auditable: who put what in front of the council, when.
- **Export formats.** Signed PDF verdict (embed the verdict CID + a verification QR resolving to the verify endpoint), ADR markdown, and JSON-LD for the civic transparency portal. The PDF pipeline reuses your existing WeasyPrint infrastructure.
- **Observability.** Per-turn telemetry artifact (tokens in/out, cache hit ratio, model, latency, cost in integer micro-USD) — content-addressed like everything else, so cost claims in public reporting are themselves verifiable. Dashboards can wait; the data capture cannot.

07 Phased Roadmap & Milestones

Milestone	Deliverable	Exit criteria
M1 — Canonical core (wk 1–2)	`conclave_core` : dag-cbor canonicalization, CIDv1, artifact model, DAG builder	Cross-checked CID test vectors vs. reference IPLD implementations; property tests on canonical-form stability
M2 — Signing & log (wk 2–4)	`conclave_crypto` : ML-DSA-65 via pqcrypto, key roles, cert chain, RFC 6962 log	Independent verifier CLI validates a full synthetic case offline; inclusion + consistency proofs green
M3 — Skill retrofit (wk 3–5)	v0.1 skill emits v0.2 artifacts via `conclave_core` CLI; retry/supersede semantics	A real deliberation run in Claude Code produces a fully verifiable DAG
M4 — Token architecture (wk 4–7)	Per-persona calls, caching layout, cascade, distillation, sparse topology, early exit, Batch mode	Measured cost on 10 benchmark deliberations ≤30% of cascaded baseline; distillation-fidelity spot audits pass
M5 — Appeals (wk 6–8)	Dissents, T1–T3 triggers, Tier A/B/C ladder, finality timers	End-to-end appeal on a seeded procedural violation, caught by public verification, reversed on appeal
M6 — conclave_server (wk 7–10)	Serverpod endpoints, Postgres artifact store, Redis session state, watch streams, budgets	Two concurrent live deliberations streamed to a test client; kill/resume mid-round leaves a clean DAG
M7 — v0.2.0 cut (wk 10–12)	Docs, verifier CLI release, signed-PDF export, one real eSiasa policy question end-to-end	Tagged release; an external party (not you) verifies the demo verdict from artifacts + pubkeys alone

v0.3+ glance: Flutter observer/verifier app; T4 stakeholder petitions wired to eSiasa identity; external witness co-signing on the transparency log; persona marketplace with third-party signed manifests; calibration-weighted voting; Swahili-language deliberation surface for the civic portal.

08 Risks, Failure Modes & Open Questions

- **Wrong-consensus convergence.** Debate literature is clear that multi-agent systems can confidently converge on wrong answers, and sparse topologies + early exit slightly increase this risk by reducing challenge surface. Mitigations already in-design: mandatory devil's-advocate seats in verdict rounds, dissent-suppression checks in distillation, and the appeals ladder as the systemic backstop. Track a metric: reversal rate on appeal is your ground-truth error signal.
- **Verifiability ≠ correctness.** The crypto layer proves *what happened*, not that what happened was *right*. Publish this distinction loudly; conflating them is the fastest way to torch credibility with exactly the institutional audiences eSiasa needs.
- **Canonicalization drift.** A dag-cbor encoding bug shipped once poisons every downstream hash. M1's cross-implementation test vectors are non-negotiable, and the canonical encoder gets frozen behind a versioned `codecVersion` field in every artifact so a future fix is a new codec version, not a silent change.
- **Pricing and API volatility.** Every number in §04 has a shelf life; cache TTLs, batch discounts, and model tiers move. The cost model lives in code (a re-runnable estimator over the telemetry schema), not in this PDF.
- **Distillation as attack surface.** Whoever controls the distiller controls what the council remembers. The distiller's model, prompt, and outputs are all content-addressed — but v0.2.0 should also randomize distiller sampling across two models and diff, escalating on divergence. Cheap insurance.
- **Open question — persona epistemic independence.** Per-persona calls fix context leakage but all personas still share one base model's priors. True diversity may require multi-provider panels (a Sovereign on one lab's model, a Justice on another's). The artifact model already supports it (`modelId` per turn); whether the quality delta justifies the operational sprawl is an empirical question for v0.3 benchmarking.
- **Open question — human arbiter throughput.** Tier C is mandatory finality, and humans do not scale. For eSiasa volumes, the screening gates (T1 materiality, T2 mechanical proof, T3 threshold) must keep Tier C arrivals rare. If they do not, the protocol needs a Tier B½ before humans drown — watch the funnel metrics from day one.

> **BOTTOM LINE**
>
> v0.2.0 is ~12 weeks of focused work, most of it in Dart you already know how to write, on cryptographic primitives you already ship. The result is a deliberation protocol nobody else has: PQ-signed, publicly verifiable, appealable, and cheap enough to run at civic scale.